



Congenital Factor XIII Deficiency: From Diagnosis to Recombinant FXIII Therapy (NovoThirteen®)



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Introduction

Congenital factor XIII (FXIII) deficiency is a rare autosomal recessive bleeding disorder characterized by impaired stabilization of the fibrin clot.

Unlike most coagulation factor deficiencies, routine coagulation tests, including prothrombin time (PT) and activated partial thromboplastin time (aPTT), are usually normal.

This particularity may lead to delayed diagnosis and expose patients to severe bleeding complications, particularly intracranial hemorrhage.

Clinical manifestations may vary from delayed umbilical cord bleeding and soft tissue hematomas to poor wound healing, recurrent pregnancy loss and severe intracranial bleeding.

This review summarizes the clinical features, diagnostic approach and therapeutic advances in congenital FXIII deficiency, with a particular focus on recombinant FXIII (NovoThirteen®).



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Objectives

2.1. Main objective :

To summarize the clinical characteristics, diagnostic approach and therapeutic management of congenital factor XIII deficiency, from diagnosis to recombinant FXIII therapy.

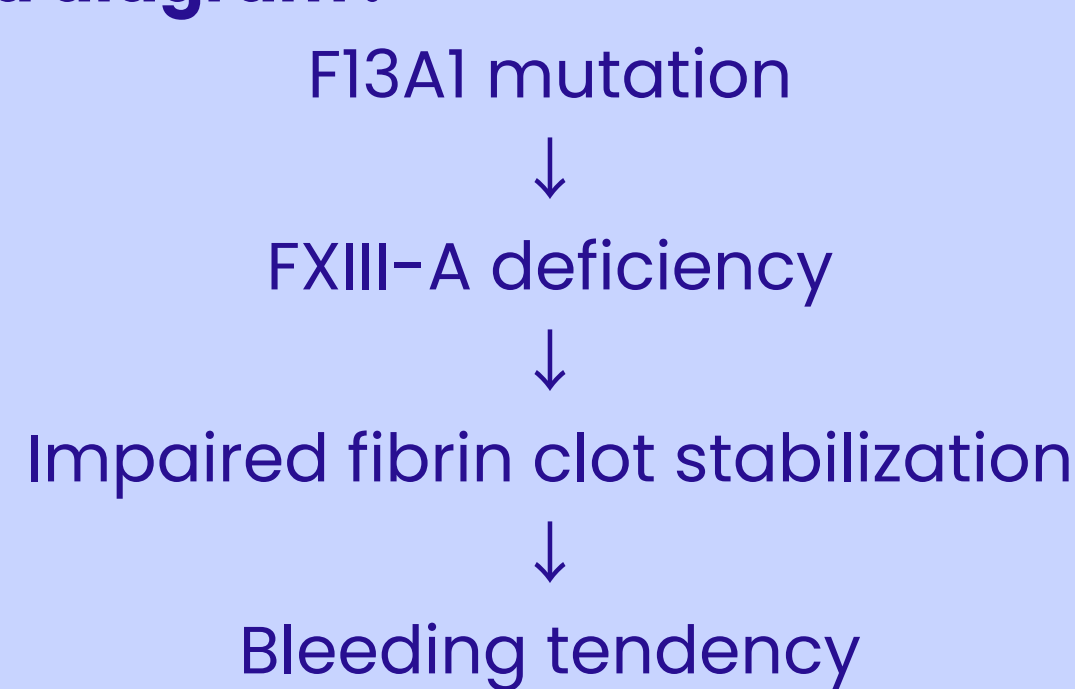
2.2. Specific objectives :

- To describe the main clinical manifestations of congenital FXIII deficiency.
- To highlight the particular laboratory features of this disorder.
- To explain the diagnostic approach in patients with normal routine coagulation tests.
- To describe the role of specific FXIII activity assays and genetic testing.
- To review current therapeutic strategies.
- To highlight the role of recombinant FXIII (NovoThirteen®) in long-term prophylaxis.

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Pathophysiology

- Most cases of congenital FXIII deficiency are caused by **mutations in the F13A1 gene**, resulting in deficiency of the **FXIII A-subunit (FXIII-A)**.
- FXIII plays a crucial role in the **stabilization of the fibrin clot**. Its deficiency therefore results in impaired clot stabilization and a tendency to delayed bleeding.
- **Suggested diagram :**



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Clinical manifestations

The clinical presentation is variable and may include:

Early manifestations :

- Delayed umbilical cord bleeding

Bleeding manifestations :

- Soft tissue hematomas
- Poor wound healing
- Intracranial hemorrhage

Obstetric manifestations :

- Recurrent pregnancy loss

Among these complications, **intracranial hemorrhage** represents one of the most severe manifestations.

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Laboratory features

One of the major characteristics of congenital FXIII deficiency is the **normality of routine coagulation tests**.

PT → usually normal

aPTT → usually normal

Therefore:

Normal PT + Normal aPTT ≠ No bleeding disorder

A patient with a significant bleeding history and normal routine coagulation tests should therefore raise suspicion of **FXIII deficiency**.

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Diagnostic approach

Because PT and aPTT are usually normal, the diagnosis cannot be established using routine coagulation tests alone.

Diagnosis relies on **specific investigations**.

1. FXIII activity assay :

Measures the functional activity of factor XIII and helps confirm the diagnosis.

2. Genetic testing :

Genetic analysis can identify mutations responsible for the congenital deficiency, particularly abnormalities involving the **F13A1 gene**.

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Recombinant FXIII: NovoThirteen®

What is NovoThirteen®?

5.1. NovoThirteen® (catridecacog) is a recombinant coagulation factor XIII-A (FXIII-A) concentrate produced by recombinant DNA technology.

Catridecacog is structurally identical to the human **FXIII-A subunit**, which contains the enzymatic activity of factor XIII. It replaces the deficient FXIII-A subunit in patients with **congenital FXIII A-subunit deficiency**.

5.2. Mechanism of action :

Normally, FXIII circulates as an **A₂B₂ tetramer**. Following activation by thrombin in the presence of calcium, FXIII-A becomes an active transglutaminase.

Activated FXIII:

Cross-links fibrin and other proteins



Increases mechanical strength of the fibrin clot



Increases resistance to fibrinolysis



Improves stabilization of the clot

In congenital FXIII A-subunit deficiency, this process is impaired. NovoThirteen® provides recombinant FXIII-A and therefore restores the deficient enzymatic activity.

5.3. Administration and prophylaxis :

NovoThirteen® is administered by **intravenous injection** as long-term prophylactic treatment.

The recommended regimen is **35 IU/kg every 28 days**

This corresponds to approximately **once monthly**, allowing regular replacement of FXIII-A and maintenance of protective FXIII levels.

The treatment is intended for long-term prophylaxis of bleeding in patients with congenital FXIII A-subunit deficiency. The EMA indication covers adults and children 6 years and older.

5.4. Therapeutic benefits :

The main objective of prophylactic recombinant FXIII therapy is to prevent spontaneous and clinically significant bleeding.

Reported benefits include:

- **Maintenance of protective FXIII levels**
- **Reduction in bleeding episodes requiring FXIII treatment**
- **Effective long-term prophylaxis**
- Prevention of severe hemorrhagic complications
- **Good safety profile**
- Improvement in patients' quality of life

In the pivotal study cited by the EMA, 41 patients aged over 6 years receiving prophylactic NovoThirteen® experienced an average of approximately **0.15 bleeding episodes requiring FXIII treatment per patient-year**.

5.5. Safety :

Generally well tolerated. Reported adverse effects include headache, leukopenia/neutropenia, limb pain, injection-site pain and anti-FXIII antibodies. Appropriate monitoring is required.

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CONCLUSION

NovoThirteen® (catridecacog), a recombinant FXIII-A concentrate, represents an important therapeutic advance in congenital FXIII A-subunit deficiency. Its 28-day prophylactic administration provides sustained FXIII replacement, reduces bleeding episodes and offers an effective and well-tolerated long-term treatment option.

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References

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